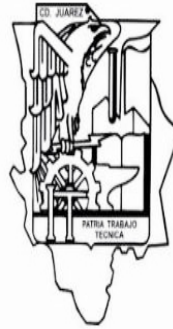




**TECNOLÓGICO
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JUÁREZ**
Forjando el futuro...

Tecnológico Nacional de México Campus Ciudad Juárez

**Ingeniería en Sistemas Computacionales
Inteligencia Artificial**

U1. Introducción a la Inteligencia Artificial

Practica#1

Grupo M

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Fecha: 25/09/2026

```
from google.colab import drive
drive.mount('/content/drive')
```

In [1]:

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
In [2]: #libreria de matplot (para graficas, colores,etc)
import matplotlib.pyplot as plt
#Libreria de seaborn
import seaborn as sns
```

Ejercicio 2

In [3]:

```
# importacion de la libreria pandas y csv de la informacion
import pandas as pd
info_comp = pd.read_csv('/content/drive/MyDrive/T-Escuela_8/Inteligencia
Artificial/Python/U1/Ejercicio 1/Forbes_2000_top_company.csv')
info_comp
```

Out [3]: Columns 1–8 of 12

	2022 Ranking	Organization Name	Industry	Country	Year Founded	CEO	Revenue (Billions)
0	1	Berkshire Hathaway	Diversified Financials	United States	1939	Warren Edward Buffett	276.09
1	2	ICBC	Banking	China	1984	Shu Gu	208.13
2	3	Saudi Arabian Oil Company (Saudi Aramco)	Oil & Gas Operations	Saudi Arabia	1933	Amin bin Hasan Al-Nasser	400.38
3	4	JPMorgan Chase	Diversified Financials	United States	2000	Jamie Dimon	124.54
4	5	China Construction Bank	Banking	China	2014	Wang Zuji	202.07
...	...	...	...	...	...	...	...
1994	1995	Aperam	Materials	Luxembourg	2010	Timoteo di Maulo	6.03
1995	1995	Shenzhen Feima International Supply Chain	Business Services & Supplies	China	1998	Shan Min Huang	0.04
1996	1997	NMDC	Materials	India	1958	Sumit Deb	3.52
1997	1997	Sichuan Changhong Electric	Consumer Durables	China	1988	Wei Li	15.72
1998	1999	Satellite Chemical	Chemicals	China	0	No Data	4.41

Columns 9–12 of 12 · first column repeated

	Profits (Billions)	Assets (Billions)	Market Value (Billions)	Total Employees
0	89.80	958.78	741.48	372000
1	54.03	5518.51	214.42	449296
2	105.36	576.04	2292.08	68493
3	42.12	3954.69	374.45	271025
4	46.89	4746.95	181.32	352621
...	...	...	...	...
1994	1.14	6.72	3.16	0
1995	1.41	0.17	1.14	306
1996	1.41	5.72	6.40	16944
1997	0.05	12.10	1.96	51841

1998	0.93	7.64	9.52	3409
------	------	------	------	------

1999 rows × 11 columns

```
In [4]: #Columnas del csv
info_comp.columns
```

```
Out[4]: Index(['2022 Ranking', 'Organization Name', 'Industry', 'Country',
              'Year Founded', 'CEO', 'Revenue (Billions)', 'Profits (Billions)',
              'Assets (Billions)', 'Market Value (Billions)', 'Total Employees'],
              dtype='object')
```

```
In [5]: #Columnas de tipo numericas
#Filtramos por valores para utilizarlos luego y saber cuales pueden
servirnos
info_comp.select_dtypes(include=['number']).sort_values('Assets
(Billions)', ascending=False)
```

	Ranking	Year Founded	Revenue (Billions)	Profits (Billions)	Assets (Billions)	Market Value (Billions)	Employees
	2	1984	208.13	54.03	5518.51	214.42	449296
4	5	2014	202.07	46.89	4746.95	181.32	352621
	8	1979	181.42	37.38	4561.05	133.38	455174
337	338	1938	100.33	22.18	4229.90	0.88	7400
12	13	1929	152.43	33.57	4192.84	117.82	306322
	...	...	...	...	...	...	...
1574	1575	0	41.81	-1.57	1.62	1.21	36992
	1905	1993	17.64	0.04	1.54	0.56	0
	1943	0	0.00	1.50	1.36	2.43	0
1745	1746	2005	0.35	0.08	0.50	34.74	517
	1995	1998	0.04	1.41	0.17	1.14	306

Out[5]:

Total

1

7

1904

1942

1995

1999 rows × 7 columns

```
In [6]: #Columnas de tipo texto
#Filtramos por texto para utilizarlos luego y saber cuales pueden servirnos
info_comp.select_dtypes(include=['object', 'string'])
```

	Organization Name			CEO
0	Berkshire Hathaway	Diversified Financials	United States	Warren Edward Buffett
	ICBC	Banking	China	Shu Gu
3	Saudi Arabian Oil Company (Saudi Aramco)	Oil & Gas Operations	Saudi Arabia	Amin bin Hasan Al-Nasser
	JPMorgan Chase	Diversified Financials	United States	Jamie Dimon
4	China Construction Bank	Banking	China	Wang Zuji
	...	...	...	...
	Aperam	Materials	Luxembourg	Timoteo di Maulo
	Shenzhen Feima International Supply Chain	Business Services & Supplies	China	Shan Min Huang
	NMDC	Materials	India	Sumit Deb
1997	Sichuan Changhong Electric	Consumer Durables	China	Wei Li
	Satellite Chemical	Chemicals	China	No Data

Out [6]:

Industry

Country

1

2

1996

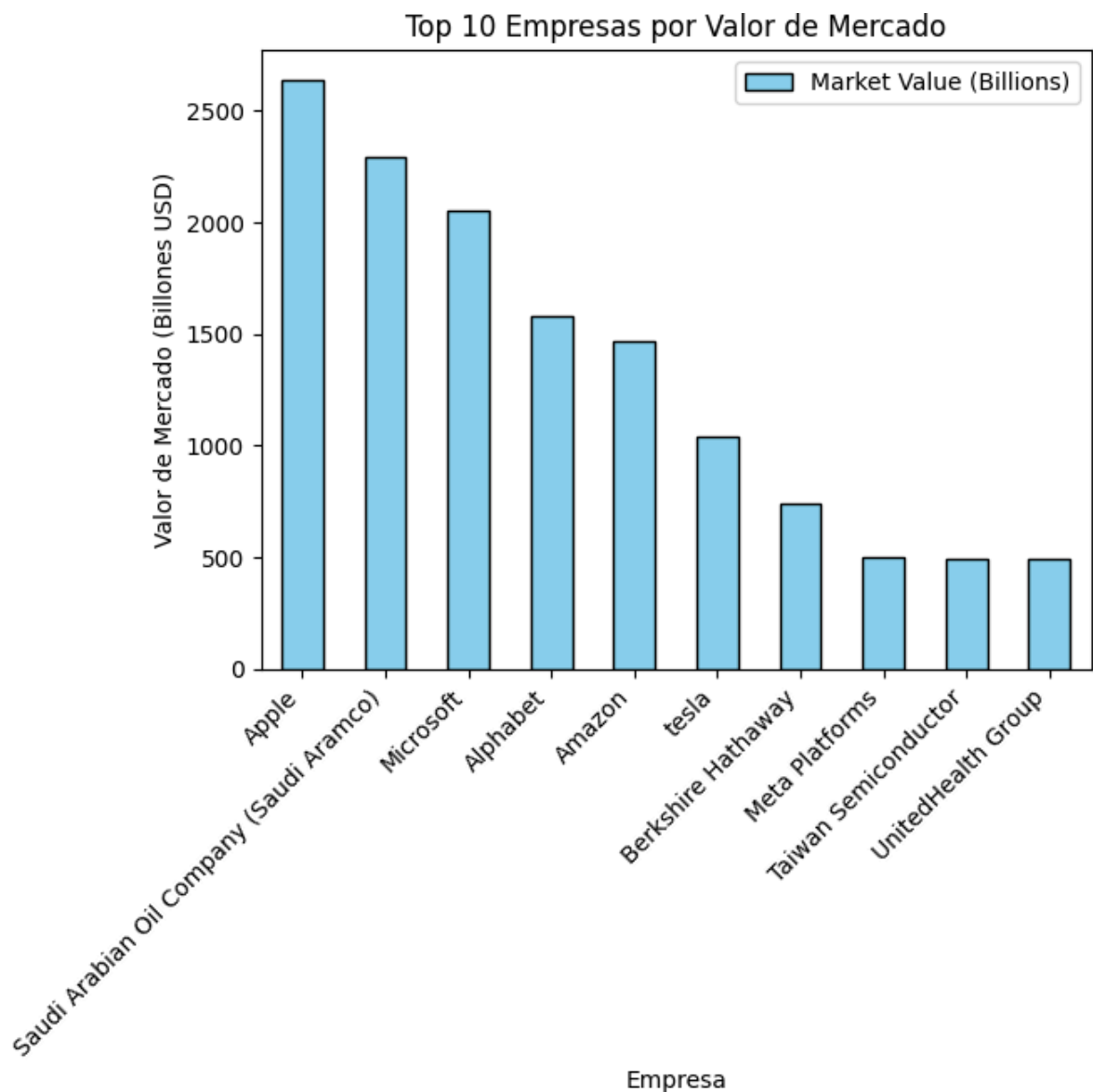
1998

1999 rows × 4 columns

Grafica de barras

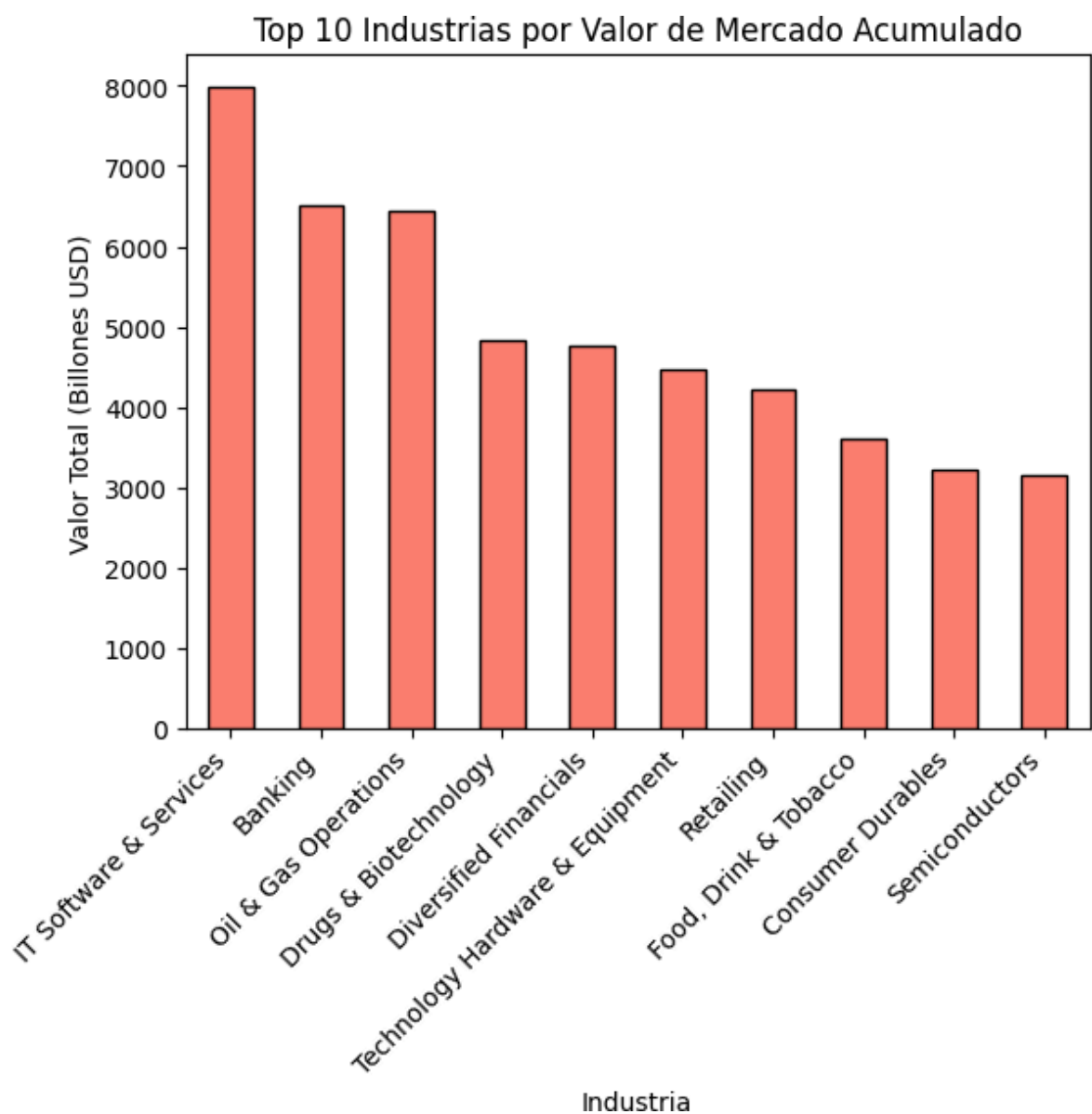
```
In [7]: # Seleccionamos las 10 empresas con mayor valor de mercado
top_empresas = info_comp.nlargest(10, 'Market Value (Billions)')

# Creamos la gráfica de barras
top_empresas.plot(
    kind='bar',
    x='Organization Name',
    y='Market Value (Billions)',
    color='skyblue',
    edgecolor='black'
)
plt.title('Top 10 Empresas por Valor de Mercado')
plt.xlabel('Empresa')
plt.ylabel('Valor de Mercado (Billones USD)')
plt.xticks(rotation=45, ha='right')
plt.show()
```



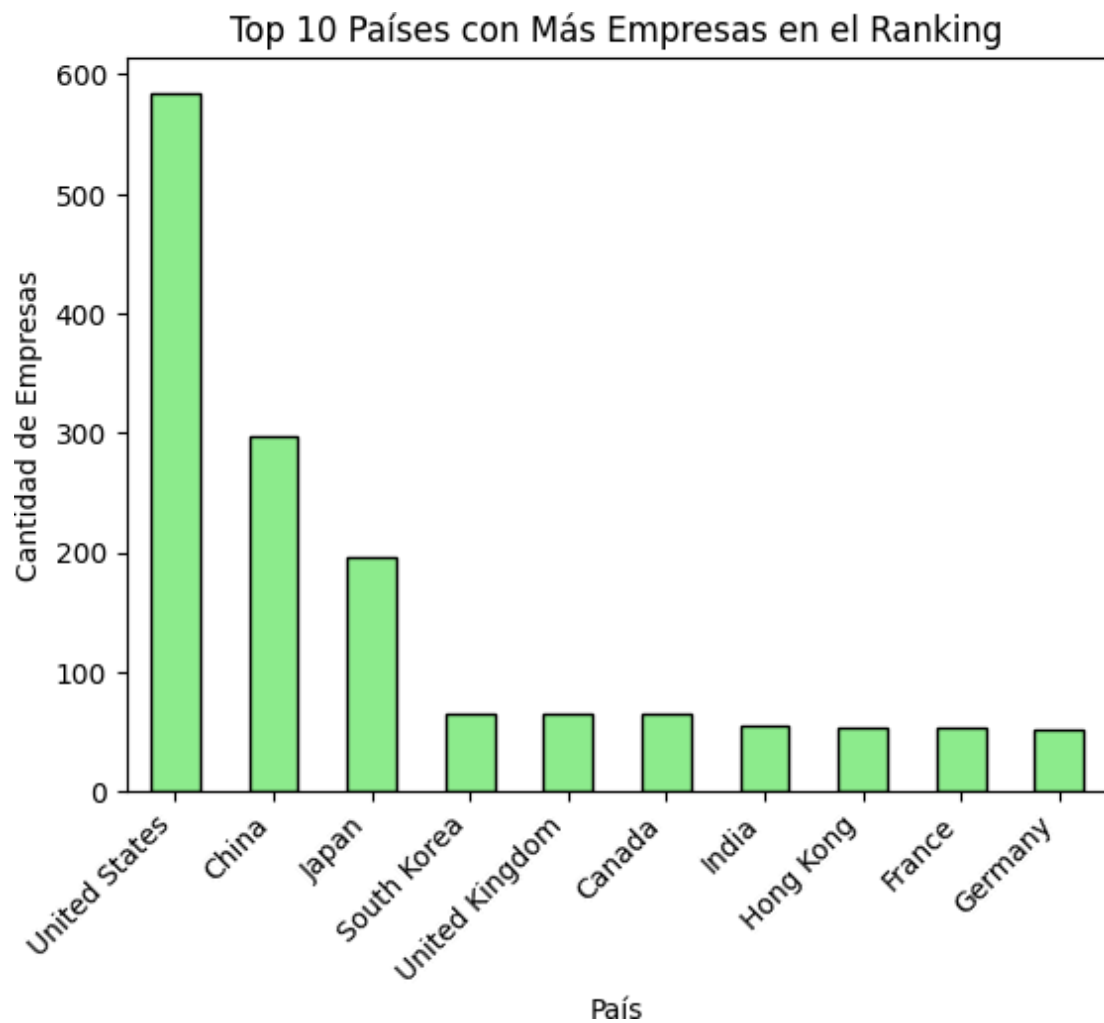
```
In [8]: # Agrupamos por industria y sumamos su valor de mercado, tomando las 10
principales
mercado_industria = (
    info_comp.groupby('Industry')['Market Value (Billions)']
    .sum()
    .nlargest(10)
)

# Gráfica
mercado_industria.plot(kind='bar', color='salmon', edgecolor='black')
plt.title('Top 10 Industrias por Valor de Mercado Acumulado')
plt.xlabel('Industria')
plt.ylabel('Valor Total (Billones USD)')
plt.xticks(rotation=45, ha='right')
plt.show()
```



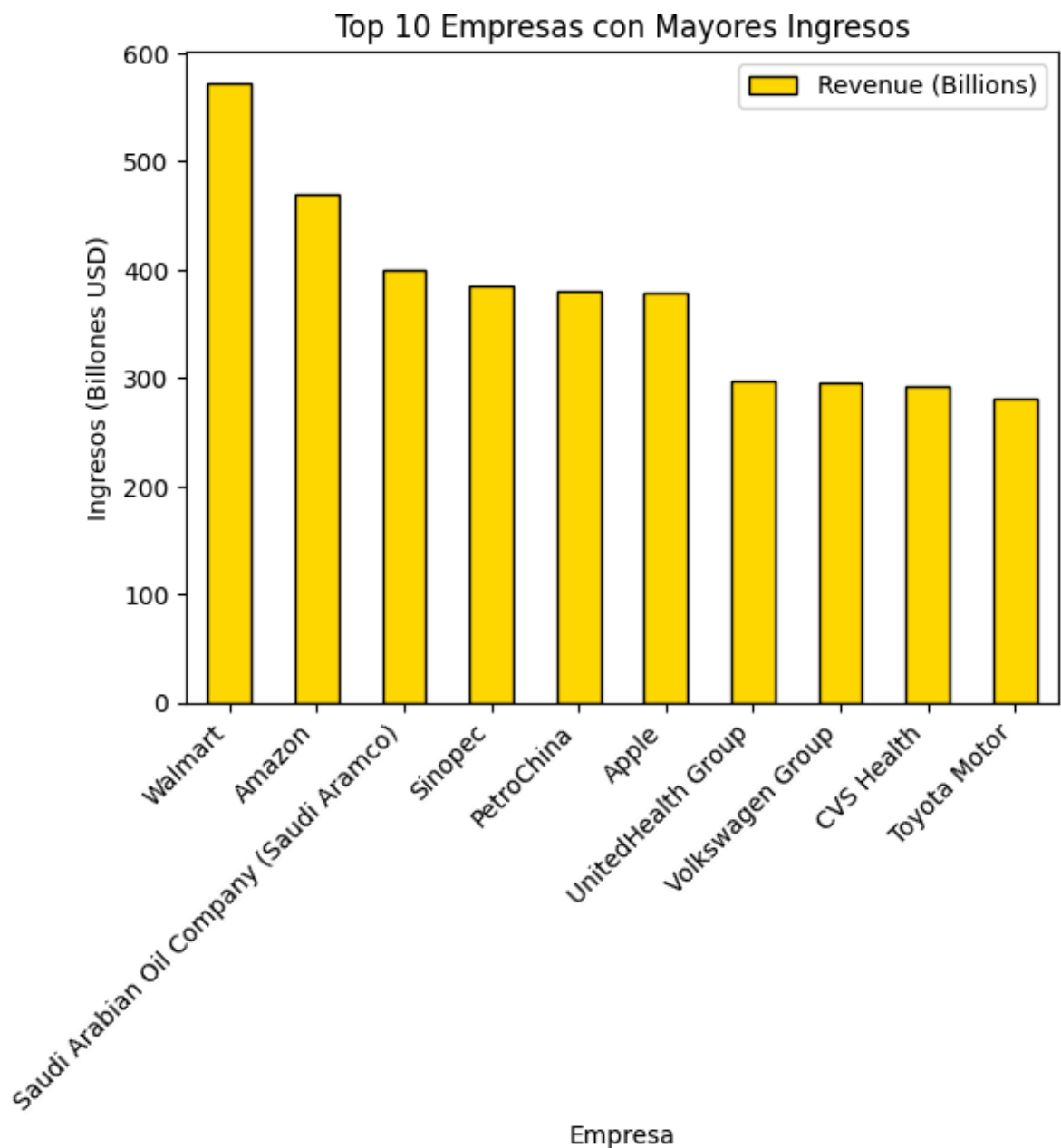
```
In [9]: # Contamos cuántas empresas hay por país y tomamos el top 10
top_paises = info_comp['Country'].value_counts().head(10)

# Gráfica
top_paises.plot(kind='bar', color='lightgreen', edgecolor='black')
plt.title('Top 10 Países con Más Empresas en el Ranking')
plt.xlabel('País')
plt.ylabel('Cantidad de Empresas')
plt.xticks(rotation=45, ha='right')
plt.show()
```



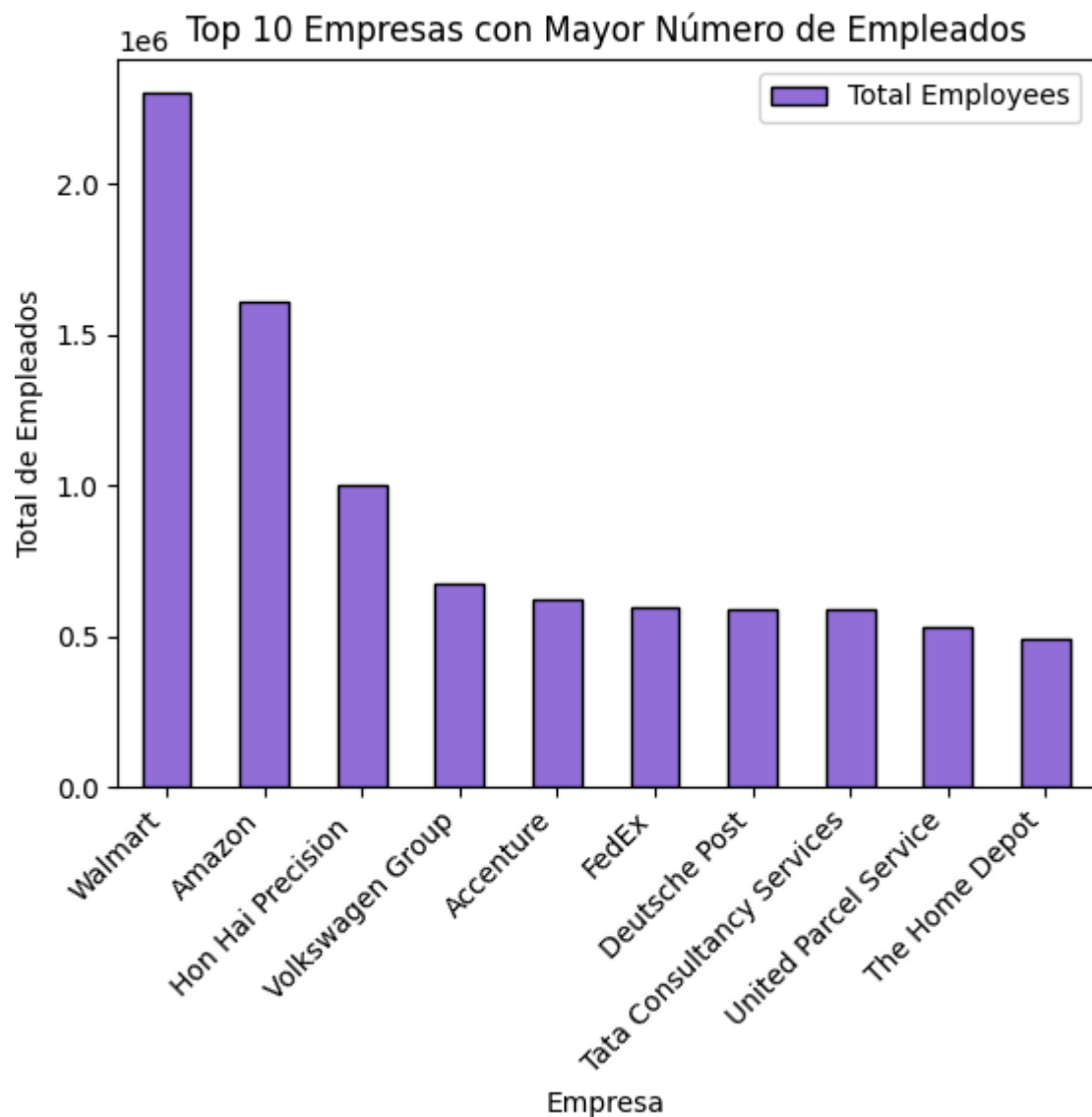
```
In [10]: # Top 10 por ingresos
top_ingresos = info_comp.nlargest(10, 'Revenue (Billions)')

# Gráfica
top_ingresos.plot(
    kind='bar',
    x='Organization Name',
    y='Revenue (Billions)',
    color='gold',
    edgecolor='black'
)
plt.title('Top 10 Empresas con Mayores Ingresos')
plt.xlabel('Empresa')
plt.ylabel('Ingresos (Billones USD)')
plt.xticks(rotation=45, ha='right')
plt.show()
```



```
In [11]: # Top 10 por número de empleados
top_empleados = info_comp.nlargest(10, 'Total Employees')

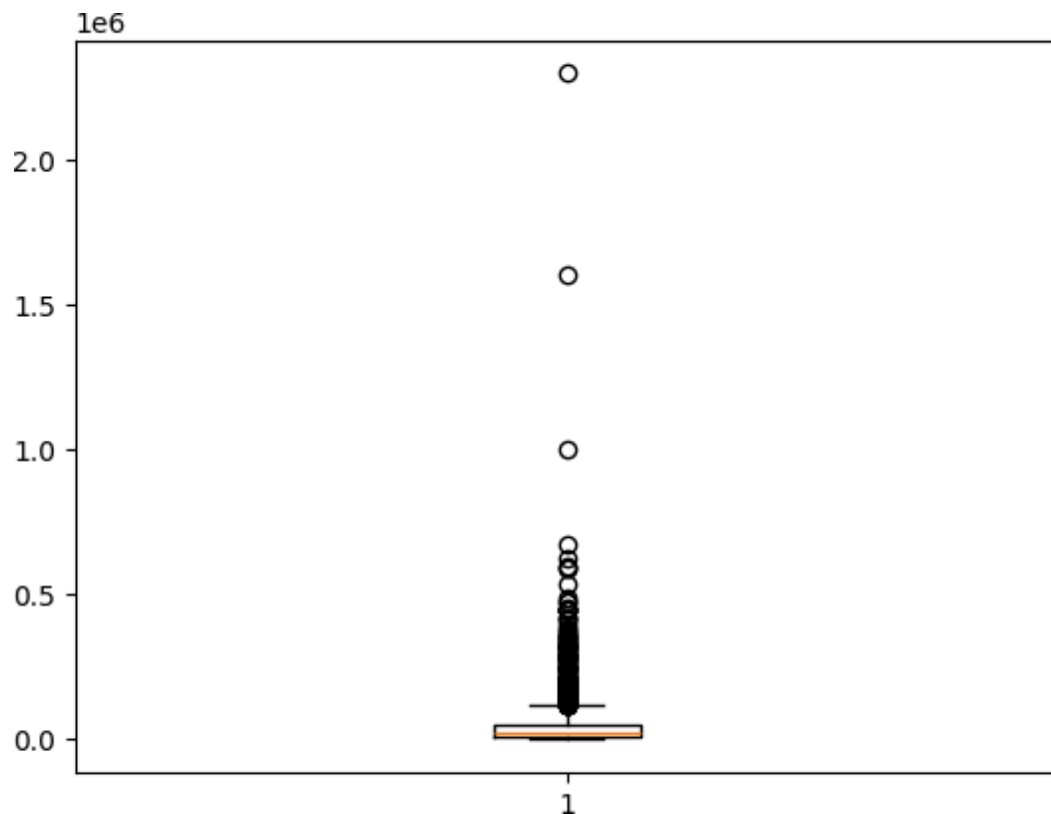
# Gráfica
top_empleados.plot(
    kind='bar',
    x='Organization Name',
    y='Total Employees',
    color='mediumpurple',
    edgecolor='black'
)
plt.title('Top 10 Empresas con Mayor Número de Empleados')
plt.xlabel('Empresa')
plt.ylabel('Total de Empleados')
plt.xticks(rotation=45, ha='right')
plt.show()
```



Grafica BoxPlot

```
In [12]: #Grafica boxplot con Matplotlib
plt.boxplot(info_comp['Total Employees'])
```

```
Out[12]: {'whiskers': <matplotlib.lines.Line2D at 0x7bfeedb2990>,
  <matplotlib.lines.Line2D at 0x7bfeedb2ad0 >,
  'caps': <matplotlib.lines.Line2D at 0x7bfeedb2c10>,
  <matplotlib.lines.Line2D at 0x7bfeedb2d50 >,
  'boxes': <matplotlib.lines.Line2D at 0x7bfeedb2850 >,
  'medians': <matplotlib.lines.Line2D at 0x7bfeedb2e90
  >, 'fliers': <matplotlib.lines.Line2D at 0x7bfeedb2fd0
  >, 'means': []}
```

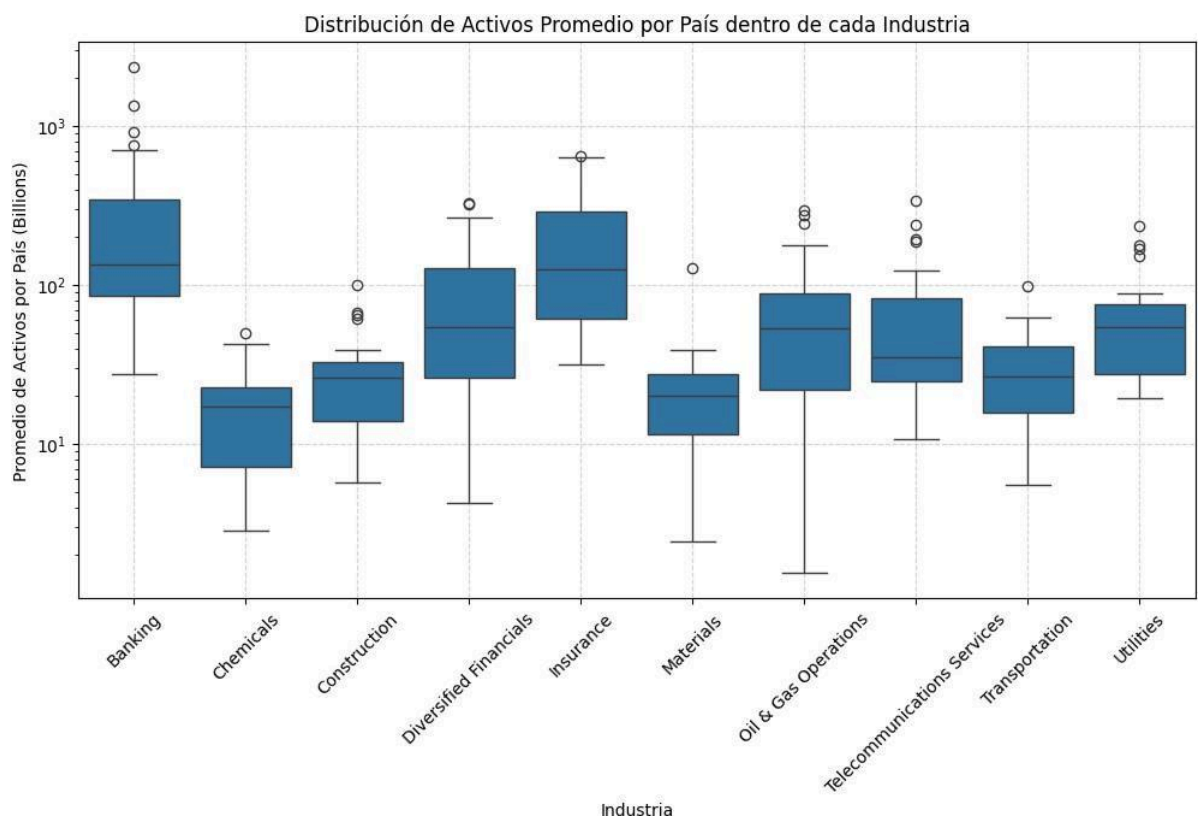


```
In [13]: # 1. CREA TABLA AGRUPADA
tabla_agrupada = info_comp[['Assets (Billions)', 'Industry',
'Country']].groupby(['Industry', 'Country']).mean().reset_index()

# 2. SELECCIONA LAS 10 INDUSTRIAS PRINCIPALES
top_industrias = tabla_agrupada['Industry'].value_counts().head(10).index
df_grafica =
tabla_agrupada[tabla_agrupada['Industry'].isin(top_industrias)]

# 3. GRAFICA
plt.figure(figsize=(12, 6))
sns.boxplot(data=df_grafica, x='Industry', y='Assets (Billions)')

plt.title('Distribución de Activos Promedio por País dentro de cada
Industria')
plt.xlabel('Industria')
plt.ylabel('Promedio de Activos por País (Billions)')
plt.xticks(rotation=45)
plt.grid(True, linestyle='--',
alpha=0.5) plt.yscale('log')
plt.show()
```



Grafica de Dispersión

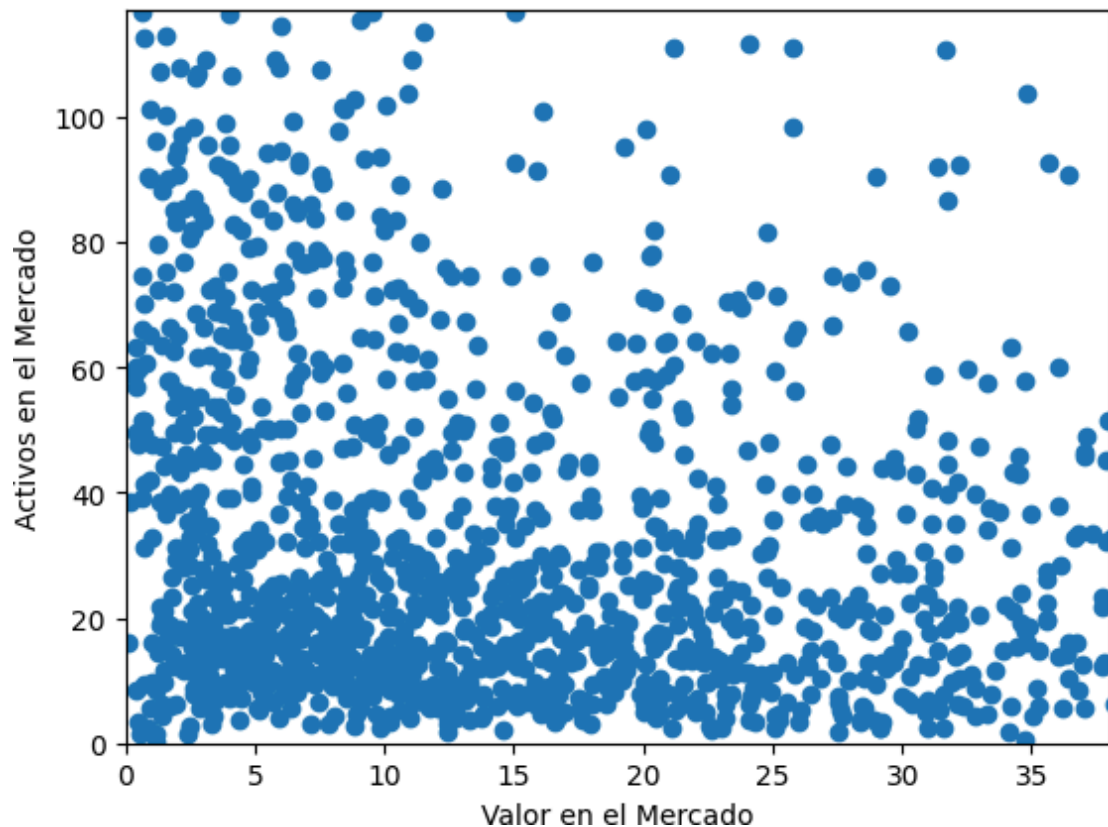
```
In [14]: print(info_comp['Market Value (Billions)'].mean().round(2))
print(info_comp['Assets (Billions)'].mean().round(2))
```

38.26

117.18

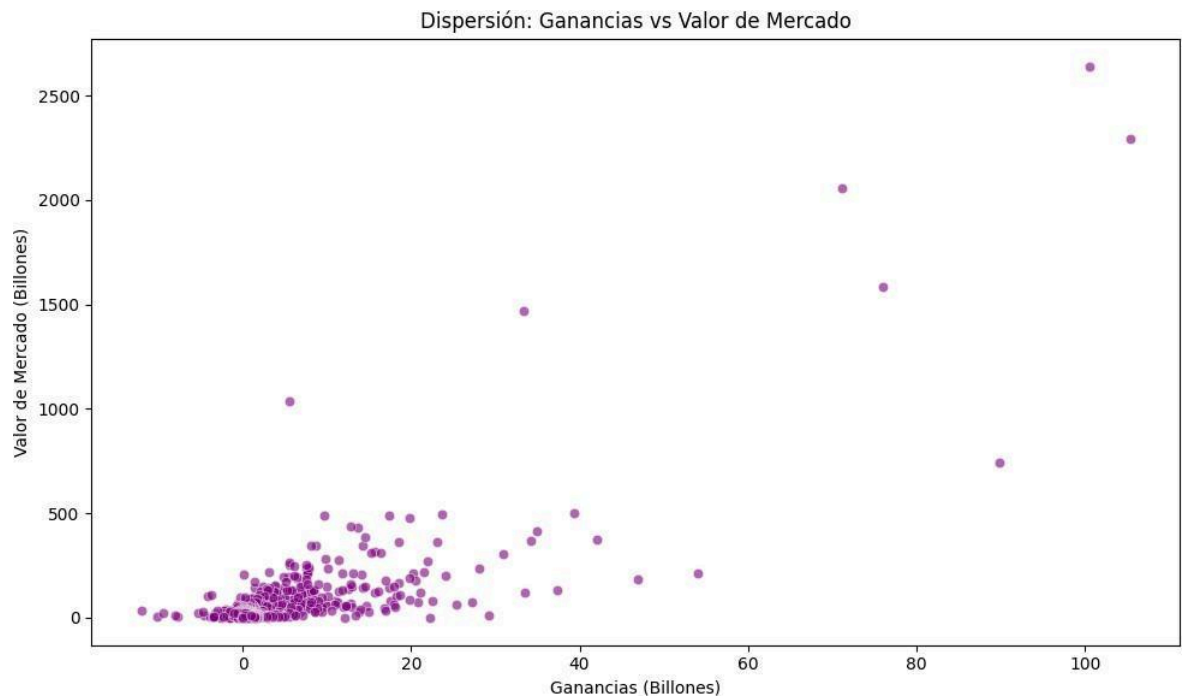
```
In [15]: #Grafica de dispersion con Matplotlib
activos = info_comp['Assets (Billions)']
valor_mercado = info_comp['Market Value
(Billions)'] plt.scatter(valor_mercado, activos)
plt.xlim(0, 38)
plt.ylim(0, 117)
plt.xlabel('Valor en el
Mercado')
plt.ylabel('Activos en el Mercado')
```

```
Out[15]: Text(0, 0.5, 'Activos en el Mercado')
```



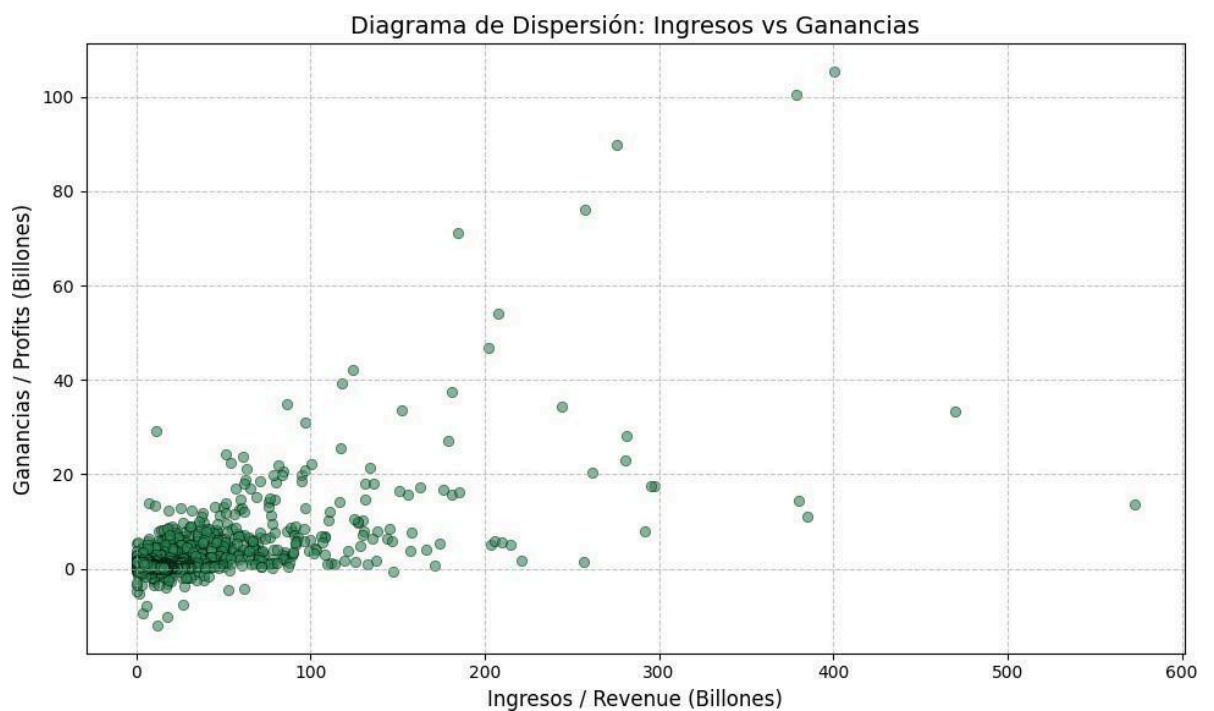
In [16]:

```
# 1. Diagrama de dispersión de Ganancias vs Valor de Mercado
plt.figure(figsize=(10, 6))
sns.scatterplot(data=info_comp, x='Profits (Billions)', y='Market Value
(Billions)', alpha=0.6, color='purple')
plt.title('Dispersión: Ganancias vs Valor de Mercado')
plt.xlabel('Ganancias (Billones)')
plt.ylabel('Valor de Mercado
(Billones)') plt.tight_layout()
plt.show()
```



```
In [17]: # 2. Diagrama de dispersión de Ingresos vs Ganancias
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=info_comp,
    x='Revenue
    (Billions) ',
    y='Profits
    (Billions) ',
    alpha=0.6,
    color='seagreen',
    edgecolor='black'
)

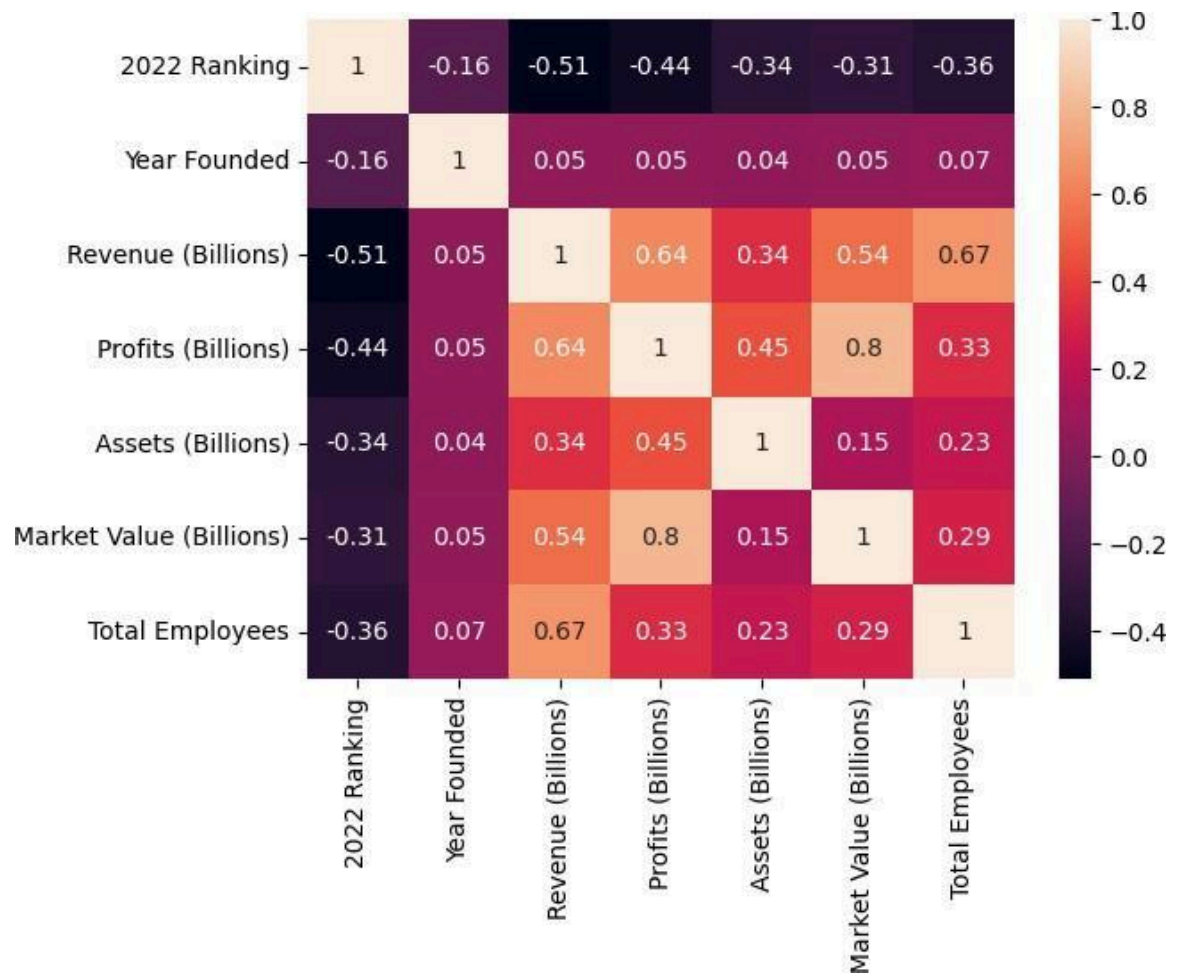
plt.title('Diagrama de Dispersión: Ingresos vs Ganancias',
fontsize=14) plt.xlabel('Ingresos / Revenue (Billones)', fontsize=12)
plt.ylabel('Ganancias / Profits (Billones)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



Grafica de Calor

```
In [18]: # Grafica de calor con pandas
sns.heatmap(round(info_comp.corr(numeric_only=True),2), annot=True)
```

Out[18]: <Axes: >



```
In [19]: # 4. Mapa de calor (Correlación entre columnas numéricas)
plt.figure(figsize=(10, 8))
correlacion = info_comp.corr(numeric_only=True)
sns.heatmap(correlacion, annot=True, cmap='coolwarm', fmt=".2f",
linewidths=0.5, vmin=-1, vmax=1)
plt.title('Mapa de Calor de Correlaciones Numéricas')
plt.tight_layout()
plt.show()
```

